

Machine Learning Engineer Assessment Report

1. Data Exploration & Anomaly Handling

- Identified missing features (market_index and quote_signal) in the December input file.
- Removed extreme anomalies based on unrealistic rate_per_mile (< \$0.50 or > \$10.00).
- Imputed missing values in weight and market_index using median strategy.

2. Cross-Validation Strategy

- Avoided random shuffling to prevent data leakage.
- Used an 80/20 Time-Series split. The training data was chronologically filtered to train on the oldest 80% and evaluate on the latest 20%.

3. Feature Engineering & Handling Inference Mismatch

- Extracted temporal features: month, day_of_week, is_weekend, and day_of_month.
- Extracted missing spatial features (pickup_lat/lon) by joining city names from the historical dataset.
- Trained a Ridge Regression model to dynamically forecast the missing market_index and quote_signal values for December, which allowed our model to output realistic, fluctuating prices across the month.

4. Modeling & Performance

- Baseline Linear Regression achieved an MAE of \$132.94.
- Advanced Gradient Boosting (HistGradientBoostingRegressor) achieved an MAE of \$111.55 (R2 = 0.85).

December Forecast Chart

